

Notes: Beneish & Whaley (JF, 1996)
An Anatomy of the “S&P Game”: The Effects of Changing the Rules

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1 Big picture

- **Question.** What happens to trading activity and stock prices when a firm is added to the S&P 500, and how did Standard & Poor's 1989 switch from immediate implementation to preannouncement change those effects?
- **Setting.** The paper studies **103 S&P 500 additions** from **January 1986 through June 1994**. The key institutional break is **October 1989**, when S&P began announcing index changes about **five business days** before they became effective.
- **Core empirical challenge.** A price increase around inclusion can come from at least three different mechanisms: **temporary price pressure**, **a more permanent downward-sloping demand effect (imperfect substitutes)**, or **improved liquidity**. The paper's job is to separate these channels and then ask how the rule change reshaped them.
- **Why the question matters.** By the late 1980s and early 1990s, index-fund assets tied to the S&P 500 had become enormous. If inclusion mechanically raises prices, then the exact announcement protocol changes who captures that price effect: index funds, arbitrageurs, or liquidity suppliers.
- **Main data.** The authors combine **intraday trade and quote data** from ISSM and TAQ with **nearby S&P 500 futures** to measure abnormal returns over close-to-open, open-to-close, and multi-day windows. This lets them decompose overnight versus intraday price adjustments.
- **Main empirical design.** The paper compares the **old policy** period (Jan. 1986–Sep. 1989) with the **new policy** period (Oct. 1989–Jun. 1994), and within the second period it further uses variation in the number of days between the announcement day and the effective day.
- **Main old-policy result.** Before preannouncement, the newly added stock earned an abnormal return of about **3.67%** from the announcement close to the next day's close. Almost all of that came **overnight**: the close-to-open abnormal return was **4.37%**, followed by a **-0.62%** open-to-close reversal the next day.
- **Main new-policy result.** After preannouncement began, the overnight jump became smaller, but the **total run-up became larger**. The abnormal return from the announcement close to the **effective-day close** was about **7.21%**. This is the institutional environment that created the "S&P game."
- **Trading-activity result.** Trading volume and trade size are largest on or right around the **effective day**, which suggests that many index funds continue to wait until the effective day to rebalance. Under the new policy, this gives **risk arbitrageurs** time to buy after the announcement and sell into index-fund demand.
- **Main credibility result.** The paper does not rely only on daily returns. It uses intraday windows, abnormal volume, abnormal trade size, quoted spreads, and direct regressions on the rule change and announcement interval length. That is what lets it separate **opening-price effects** from **multi-day run-ups**.
- **Economic interpretation.** The evidence points to a **mostly permanent inclusion effect** plus a **temporary arbitrage-driven component** under the new rule. The rule change reduced

opening imbalances but increased the total premium index funds paid.

- **Main takeaway.** S&P's attempt to smooth implementation had an unintended side effect: it created profitable room for front-running by arbitrageurs. In equilibrium, the policy shift moved value away from passive indexers and toward traders who could anticipate their demand.

2 Data and measurement (institutional context and key objects)

2.1 Why S&P 500 additions matter in the first place

- The background is the rise of **indexing** after the Sharpe–Lintner CAPM. If investors track the market portfolio passively, the S&P 500 becomes a natural benchmark and implementation vehicle.
- The paper emphasizes how large indexed wealth had become by the sample period.
- The **Vanguard Index Trust–500 Portfolio** grew from about **\$14 million in 1976** to about **\$9.356 billion in 1994**.
- Vanguard's **Institutional Index Fund** held about **\$3.27 billion** at the end of 1994, and **Wells Fargo Nikko** managed about **\$64.61 billion** in S&P funds.
- One industry estimate put **all S&P 500 indexed assets**, public plus private, at roughly **10% of total index market value**, or about **\$335 billion** at year-end 1994.
- The Microsoft addition illustrates the scale. Because Microsoft was roughly **1%** of the index at the time, every **\$100 billion** pegged to the S&P 500 implied roughly **\$1 billion** in Microsoft purchases.

Interpretation. The paper's event-study effects are economically important only because index-fund demand is huge. Without the growth of indexing, the rule change would have mattered much less.

2.2 How S&P changes index composition

- The S&P 500 is a **market-value-weighted** index of **500 stocks** from the NYSE, AMEX, and Nasdaq National Market System.
- **Additions occur only when a stock must be removed.**
- The most common reason for deletion is a **merger or acquisition**. Other reasons include corporate restructuring, bankruptcy, recapitalization, or failing to meet index criteria.
- Replacement firms come from a **secret candidate pool** approved by the S&P Index Committee.
- Selection criteria include **industry representation**, **firm size**, **breadth of ownership**, **trading volume**, and **financial soundness**.

Interpretation. Additions are not random. S&P is trying to preserve an investable benchmark, which means additions tend to be large, liquid, and important firms. That matters when interpreting the permanence of the price effect.

2.3 Old announcement policy versus new announcement policy

- Under the **old policy** (before October 1989), S&P announced the change **after the market close**, and the change became effective by the **following morning's open**. There was essentially no time for outsiders to trade between announcement and implementation.
- Under the **new policy** (starting October 1989), S&P still announced after the close, but it also stated the **future effective date**, usually about **five business days later**.
- S&P's stated motive was to **ease order imbalances** that had arisen when index funds tried to buy all at once the morning after the announcement.
- The interval was usually five days, but not always. It could be shorter if the removal had to happen quickly, and longer in unusual cases such as **Microsoft's 16-day interval** in 1994.

Interpretation. This institutional change is the paper's central quasi-experiment. It does not change *what* gets added to the index; it changes *when others can trade on the information*. That is what creates the S&P game.

2.4 Sample construction

- The raw universe is **177 additions** to the S&P 500 between **January 1986 and June 1994**.
- The authors drop **45 pre-1993 Nasdaq additions** because intraday trade-and-quote data are unavailable on ISSM for those firms.
- They also drop **29 additions** whose index-change announcements are contaminated by nearby firm-specific news.
- The contaminants include earnings or earnings forecasts (12 cases), dividend announcements (5), acquisitions or reorganizations (2), spin-offs (2), multiple announcements (2), and several one-off corporate events.
- The final sample contains **103 additions**.
- Of these, **33 additions** occur under the new preannouncement regime.
- The paper does *not* study deletions because deletions are often tied to firm-specific events and, in many cases, the removed stock no longer trades in a comparable way after the event.

Interpretation. The paper deliberately sacrifices sample size to get cleaner event windows. That is important because the whole identification strategy is built on isolating index-composition news from other corporate news.

2.5 Timing variation within the new-policy sample

- For the **33 post-October 1989 additions**, the number of trading days between the announcement day and the effective day ranges from **1** to **16**.
- The **mode is 5 days**, which matches S&P's stated policy.
- The **mean is 4.15 days**.
- The main outliers are **Dean Witter Discover** in 1993 with **10 days** and **Microsoft** in 1994 with **16 days**.

Interpretation. This cross-sectional variation lets the paper test whether a longer preannouncement interval merely smoothes the price response or instead gives arbitrageurs more time to front-run.

2.6 Market data sources

- For **1986–1992**, transaction prices, volumes, and bid/ask quotes come from **ISSM**.
- For **January 1993–September 1994**, the same variables come from the NYSE's **TAQ** database.
- Dividend and split information are checked against the **CRSP Daily Master File**.
- Returns on the market proxy come from the **nearby S&P 500 futures contract**, using data from the **CME** through 1993 and **Tick Data, Inc.** in 1994.
- The paper uses only trades and quotes from the stock's **primary exchange**.

Interpretation. The primary-exchange restriction is meant to keep “announcement-day” trading activity interpretable. If S&P announces only after the main exchange close, abnormal same-day activity on that exchange points to leakage or speculation rather than mechanical reaction to a public release.

2.7 Main variables and normalization

- **Abnormal trading volume** is daily volume divided by average daily volume over the **60 trading days before the announcement**.
- **Abnormal trade size** is average shares per trade on a given day divided by the average of daily average trade size over the same **60-day preannouncement window**.
- **Abnormal quoted bid/ask spread** is the day's average quoted spread divided by the mean daily spread over the **60 days before the announcement**.
- **Abnormal return** is the stock return over the event window *less* the return on nearby **S&P 500 futures** over the same interval.

Interpretation. All three trading-activity measures are relative measures. A value above one means “more than normal” for volume or trade size; a value below one means “narrower than normal” for spreads.

2.8 Why the paper uses S&P 500 futures rather than the cash index

- The paper wants to separate **overnight** returns from **intraday** returns.
- The reported opening value of the **cash S&P 500 index** is a poor benchmark for overnight returns because it is built from many component stocks whose first trades can occur after 9:30 and may initially reflect stale prices.
- The opening price of the **nearby S&P 500 futures** contract is the price of one security traded almost immediately after 9:30, which makes it a much cleaner benchmark for close-to-open event windows.

Interpretation. This choice is one of the paper’s main methodological contributions. Without it, the overnight-versus-intraday decomposition would be much noisier and much less informative.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing hypotheses for the inclusion effect

The paper frames the inclusion effect using three main hypotheses:

- **Price pressure (temporary).** Index funds create a one-time buy-order imbalance, pushing price above fundamental value; later reversal follows when liquidity suppliers unwind.
- **Imperfect substitutes (permanent).** Stocks inside and outside the S&P 500 are not perfect substitutes once indexed demand matters, so the demand shift can permanently raise price.
- **Liquidity (permanent).** If inclusion permanently narrows bid/ask spreads, the present value of future trading costs falls, permanently increasing price.

Under the new preannouncement rule, the paper adds a fourth mechanism:

- **Risk-arbitrage front-running.** Arbitrageurs buy after the announcement and try to sell to index funds on the effective day.

Interpretation. The logic of the paper is: use volume, trade size, spreads, and timing to decide which combination of these hypotheses best fits the data.

3.2 Futures-based abnormal return measure

For any event window $[t_0, t_1]$, the paper measures abnormal return as the return on the added stock minus the return on nearby S&P 500 futures over the same interval:

$$AR_{i,[t_0,t_1]} \approx R_{i,[t_0,t_1]}^{\text{stock}} - R_{m,[t_0,t_1]}^{\text{fut}}. \quad (1)$$

Equivalently, the paper interprets this as the return on a feasible hedged strategy:

- buy the newly added stock,
- short an appropriate quantity of S&P 500 futures,
- close both positions at the end of the window.

Interpretation. This is not just a statistical abnormal return. It is also the payoff to an implementable trading strategy, which is exactly what matters for the S&P game.

3.3 Abnormal trading-activity measures

The trading-activity ratios are of the form

$$AVOL_{i,d} = \frac{Vol_{i,d}}{\frac{1}{60} \sum_{s=-60}^{-1} Vol_{i,s}}, \quad (2)$$

$$ASIZE_{i,d} = \frac{TradeSize_{i,d}}{\frac{1}{60} \sum_{s=-60}^{-1} TradeSize_{i,s}}, \quad (3)$$

$$ASPREAD_{i,d} = \frac{Spread_{i,d}}{\frac{1}{60} \sum_{s=-60}^{-1} Spread_{i,s}}. \quad (4)$$

Interpretation. These ratios reveal *when* index funds seem to be rebalancing. If passive funds minimize tracking error, the biggest abnormal volume and trade size should appear on the **effective day**.

3.4 Event-window decomposition

The abnormal returns are computed over several windows:

- **close-to-close** windows such as announcement close to effective-day close or effective day plus 10, 20, 40, and 60 closes,
- **close-to-open** overnight windows,
- **open-to-close** intraday windows,
- and special intervals such as the open on the day after the announcement to the close on the effective day.

Within the post-1989 sample, the paper estimates these separately for:

- cases with exactly **one trading day** between announcement and effectiveness,
- and cases with **more than one trading day** between the two.

Interpretation. This decomposition is what lets the authors say whether the price change happens immediately, gradually, or partly reverses after index funds finish buying.

3.5 Direct test of the rule change

To test whether the new policy reduced the opening-price jump, the paper estimates

$$ABRET_i^{C \rightarrow O} = \alpha + \beta DUMMY_i + \varepsilon_i, \quad (5)$$

where:

- $ABRET_i^{C \rightarrow O}$ is the abnormal return from the announcement-day close to the following morning's open,
- $DUMMY_i = 1$ for additions after **October 1989**, and 0 otherwise.

The estimated regression is

$$ABRET_i^{C \rightarrow O} = 4.372 - 1.254 DUMMY_i. \quad (6)$$

Interpretation. The negative coefficient means the preannouncement rule did reduce the immediate overnight jump, although only **marginally significantly**. So the rule worked in the narrow sense of easing the next-morning imbalance.

3.6 Does a longer interval help or hurt?

Within the post-1989 sample, the paper next regresses returns on the number of trading days between announcement and effectiveness:

$$ABRET_i^{C \rightarrow O} = \alpha + \beta DAYS_i + \varepsilon_i, \quad (7)$$

$$ABRET_i^{O \rightarrow C} = \alpha + \beta DAYS_i + \varepsilon_i, \quad (8)$$

where $ABRET_i^{O \rightarrow C}$ is the abnormal return from the open on the day after the announcement to the close on the effective day.

The estimated equations are

$$ABRET_i^{C \rightarrow O} = 4.623 - 0.3625 DAYS_i, \quad (9)$$

$$ABRET_i^{O \rightarrow C} = 1.459 + 0.6147 DAYS_i. \quad (10)$$

Interpretation. More days *reduce* the overnight jump but *increase* the subsequent run-up. That pattern is exactly what one would expect if a longer interval gives arbitrageurs more time to play the S&P game.

3.7 Trend in the profitability of the S&P game

To see whether the open-to-close component is shrinking over time as the market learns, the paper estimates

$$ABRET_i^{O \rightarrow C} = \alpha + \beta DAY S_i + \gamma TREND_i + \varepsilon_i, \quad (11)$$

with estimated coefficients

$$ABRET_i^{O \rightarrow C} = 8.239 + 0.8307 DAY S_i - 0.0032 TREND_i. \quad (12)$$

The adjusted R^2 is about **15%**.

Interpretation. The negative time trend is suggestive rather than strong. The authors' economic point is that, as awareness spreads and indexers adapt, the profitable gradual run-up should eventually shrink or disappear.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the effect of the rule change

The paper compares the **same type of event**—S&P 500 additions—before and after a clearly dated change in S&P's implementation policy.

Interpretation. This is stronger than comparing additions to unrelated events. The institutional object is constant; only the announcement rule changes.

4.2 Why intraday decomposition is essential

Daily event studies cannot tell whether the postannouncement return is realized:

- immediately overnight,
- gradually during the next trading day,
- or over several days before the effective date.

Interpretation. This distinction is crucial. If the return is entirely overnight, the price effect is hard to trade on after the public announcement. If it unfolds gradually before the effective day, then a genuine trading strategy exists.

4.3 Why the futures benchmark improves measurement

Using nearby S&P 500 futures instead of the reported cash index reduces measurement error in overnight abnormal returns.

Interpretation. The paper’s strongest empirical claim—that the old-policy premium was mainly overnight but the new-policy premium unfolded over several days—depends directly on this measurement choice.

4.4 How the paper infers index-fund timing

The paper cannot observe each index fund’s order directly. Instead, it infers timing from:

- abnormal **trading volume**,
- abnormal **trade size**,
- and the exact timing of **return run-ups and reversals**.

Interpretation. This is a reasonable revealed-preference strategy. If the biggest abnormal trade size shows up on the effective day, that is strong indirect evidence that many passive funds still wait until then.

4.5 How the paper handles confounding news

The authors remove additions whose announcement windows overlap with obvious firm-specific information such as earnings news, dividend news, reorganizations, and other major corporate events.

Interpretation. This is important because the paper wants the return to be about **index inclusion**, not about the firm itself.

4.6 Why the liquidity hypothesis is not the whole story

A permanent liquidity effect would require a **persistent** fall in trading costs. But the quoted-spread evidence shows only **temporary** reductions in spreads.

Interpretation. This does not prove liquidity is irrelevant, but it does rule out the cleanest “permanent price increase because spreads permanently narrowed” story.

4.7 What the evidence can and cannot distinguish

What the paper does well:

- identifies a large and precisely timed return effect,
- shows how the rule change reshaped the timing of that effect,
- and documents both permanent and temporary components.

What the paper cannot fully pin down:

- whether the permanent component is *exactly* an imperfect-substitutes effect or some broader recognition/demand effect,

- the precise identity and size of each arbitrageur's trades,
- or long-horizon postinclusion drift with complete confidence, because those estimates become sensitive to outliers.

Interpretation. The paper's strongest claims are about **timing**, **trading behavior**, and the **unintended effects of the announcement rule**, not about a fully structural decomposition of demand curves.

4.8 Why the outlier discussion matters

At long horizons, the abnormal-return averages become noisier. The paper shows that removing stocks with a **daily posteffective return of 10% or more in absolute value** changes the 60-day averages meaningfully.

Interpretation. That is a warning not to overread the exact 40- or 60-day numbers. The economically important fact is already visible by the **effective day** and within the **next 10 to 20 trading days**.

5 Tables and figures

5.1 Figure 1: Growth of S&P 500 index-fund assets

Read.

- The Vanguard 500 Portfolio grows from about **\$14 million in 1976** to about **\$9.356 billion in 1994**.
- The same fund's assets increase by about a factor of **7.7** from **1983 to 1989**.
- From **1989 to 1994**, assets rise by more than **400%**.

Economic message. The inclusion effect should be stronger in the late 1980s and early 1990s than in earlier papers simply because indexed wealth is dramatically larger.

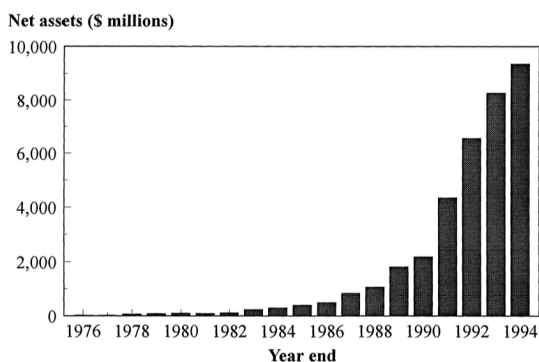


Figure 1. Asset value in millions of dollars of the Vanguard Index Trust-500 Portfolio at the end of December of each year from 1976 through 1994.

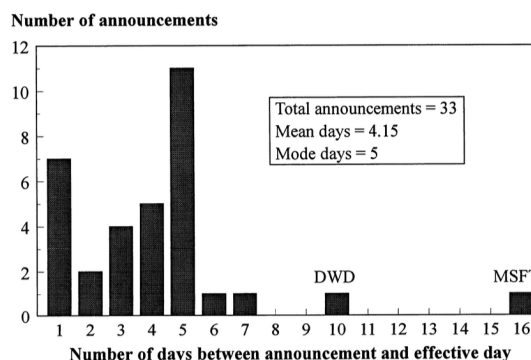


Figure 2. Frequency distribution of the number of trading days between the announcement day and the effective day for the sample of 33 S&P 500 additions during the period October 1989 through June 1994.

5.2 Figure 2: Number of trading days between announcement and effective day

Read.

- The post-1989 sample contains **33 additions**.
- The number of days between announcement and effectiveness ranges from **1** to **16**.
- The **mean is 4.15** and the **mode is 5**.
- Two visible outliers are **Dean Witter Discover** at **10 days** and **Microsoft** at **16 days**.

Economic message. The new rule usually gave the market five days, but meaningful variation in that interval lets the paper test whether more time smoothes prices or instead encourages front-running.

5.3 Table 1, Panel A: Abnormal trading volume

Read.

- Under the **old policy**, abnormal trading volume on the day after the announcement is about **7.311 times normal**.
- Under the **new policy**, the day-after-announcement volume is smaller on average: about **5.033 times normal** when weighting the one-day and more-than-one-day cases.
- But with more than one day between the announcement and effective dates, volume is still about **3.484 times normal** on the first day after the announcement, **7.212 times normal** on the days in between, and **10.565 times normal** on the **effective day** itself.
- Summing the more-than-one-day case, total abnormal volume from the day after the announcement through the effective day is about **21.261 times normal**.
- Even after the effective day, volume stays above its old baseline. For example, in the first ten days after effectiveness it is about **1.688 times normal** under the old rule and **1.968 times normal** under the new rule.

Economic message. The new policy did not eliminate abnormal trading; it *redistributed* it across days. Volume patterns strongly suggest that many index funds wait until the effective day, while arbitrageurs trade before then.

5.4 Table 1, Panels B and C: Trade size and quoted spreads

Read.

- Under the old policy, abnormal trade size on the day after the announcement is about **3.157 times normal**.

Table I
Abnormal Trading Volume, Abnormal Trade Size, and Abnormal Quoted Bid/Ask Spread of Stocks Added to the S&P 500 Index in the Days Surrounding the Announcement and Effective Days

Abnormal trading volume is defined as the stock's daily trading volume divided by the average trading volume of the stock during the 60 trading days immediately preceding the announcement. Abnormal trade size is defined as the ratio of the stock's average daily trade size (i.e., daily trading volume divided by number of trades) to the mean of the average daily trade size over the 60 trading days preceding the announcement. Abnormal quoted bid/ask spread is defined as the stock's average daily bid/ask quote divided by the mean of the average daily bid/ask quotes of the stock during the 60 trading days preceding the announcement. In the first subperiod, the announcement day and the effective day are the same. In the second subperiod, the announcement day precedes the effective day by at least one trading day. The second subperiod's postannouncement/effective day interval results are partitioned by whether one day or more separate the announcement and effective days in order to distinguish between the day following the announcement day and the effective day.

Interval		First Subperiod Jan/86–Sep/89			Second Subperiod Oct/89–Jun/94		
From	Until	<i>n</i>	Mean	<i>t</i> -ratio	<i>n</i>	Mean	<i>t</i> -ratio
Panel A: Abnormal Trading Volume							
ann. day		65	0.990	−0.11	33	1.399	2.12
Ann. day and eff. day are same.							
ann. day +1/eff. day +1		70	7.311	9.52			
One day between ann. and eff. days							
ann. day +1/eff. day					7	10.788	2.95
More than one day between ann. and eff. days							
ann. day +1					26	3.484	5.24
between ann. day +1 and eff. day					24	7.212	5.12
eff. day					26	10.565	5.71
eff. day +1							
eff. day +2	eff. day +10	70	1.688	6.50	33	3.460	6.11
eff. day +10	eff. day +20	70	1.632	3.74	33	1.616	3.55
eff. day +20	eff. day +40	69	1.301	3.62	33	1.734	2.75
eff. day +40	eff. day +60	68	1.324	2.76	33	1.553	3.44
Panel B: Abnormal Trade Size							
ann. day		65	0.937	−0.92	33	1.235	1.85
Ann. day and eff. day are same.							
ann. day +1/eff. day +1		70	3.157	10.56			
One day between ann. and eff. days							
ann. day +1/eff. day					7	2.440	3.60

Table I—Continued

Interval		First Subperiod Jan/86–Sep/89			Second Subperiod Oct/89–Jun/94		
From	Until	<i>n</i>	Mean	<i>t</i> -ratio	<i>n</i>	Mean	<i>t</i> -ratio
Panel B: Continued							
More than one day between ann. and eff. days							
ann. day +1					26	1.614	4.13
between ann. day +1 and eff. day					24	1.715	2.15
eff. day					26	4.146	7.15
eff. day +1							
eff. day +2	eff. day +10	70	1.269	5.03	33	1.571	4.60
eff. day +10	eff. day +20	70	1.092	2.06	33	1.035	0.60
eff. day +20	eff. day +40	69	0.943	−1.42	33	1.009	0.14
eff. day +40	eff. day +60	68	0.947	−1.12	33	0.909	−1.70
Panel C: Abnormal Bid/Ask Spread							
ann. day		65	0.966	−1.55	30	1.021	1.29
Ann. day and eff. day are same.							
ann. day +1/eff. day +1		70	0.871	−8.65			
One day between ann. and eff. days							
ann. day +1/eff. day					7	0.916	−1.81
More than one day ann. and eff. days							
ann. day +1					23	0.978	−0.96
between ann. day +1 and eff. day					21	1.005	0.12
eff. day					22	0.950	−1.17
eff. day +1							
eff. day +2	eff. day +10	70	0.907	−7.20	30	0.893	−3.29
eff. day +10	eff. day +20	70	0.980	−1.33	30	0.951	−1.63
eff. day +20	eff. day +40	69	1.008	0.43	30	0.973	−1.06
eff. day +40	eff. day +60	68	1.018	1.01	30	0.978	−0.79
					30	1.036	0.91

- Under the new policy with more than one day between announcement and effectiveness, abnormal trade size is about **1.614** on the first day after the announcement, **1.715** between the announcement and effective day, and peaks at about **4.146** on the **effective day**.
- Quoted bid/ask spreads narrow temporarily but not permanently. Under the old policy the spread ratio is about **0.871** on the day after the announcement; under the new policy the cleanest significant narrowing is around **0.893** on the day after the effective day.
- Outside those short windows, spread ratios are generally close to **1.0**.

Economic message. Trade size confirms that the largest trades occur when the change becomes effective. The spread evidence is too temporary to support a pure permanent-liquidity explanation for the price increase.

5.5 Table 2, Panel A: Close-to-close abnormal returns

Read.

- Across the full sample, the abnormal return from the announcement close to the close of the first post-effective trading day is about **4.388%**.
- Under the **old policy**, the corresponding return is about **3.674%**.
- Under the **new policy**, the abnormal return from the announcement close to the **effective-day close** is about **7.211%**.

- After the new-policy stock actually enters the index, some of that run-up reverses: the return measured from the announcement close to the close **one day after** effectiveness is about **5.903%**, and by **10 days after** effectiveness it is still about **4.928%**.
- In the old-policy period, the abnormal return remains about **4.638%** even **10 days** later, which is hard to reconcile with a purely temporary price-pressure story.

Economic message. Most of the inclusion effect appears permanent, but the new policy adds a temporary run-up before the effective day that later partially reverses.

Table II
Average Abnormal Returns to Stocks Added to the S&P 500 Index in the Days Surrounding the Announcement and Effective Days

In the first subperiod, the announcement day and the effective day are the same. In the second subperiod, the announcement day precedes the effective day by at least one trading day. The second subperiod postannouncement/preeffective day results are reported separately for the index additions with only one day and with more than one day between the announcement and effective days. Abnormal return is defined as the stock return over the indicated interval less the nearby S&P 500 futures return over the corresponding interval.

Return Measurement Interval		Full Sample Jan/86–June/94		First Subperiod Jan/86–Sep/89		Second Subperiod Oct/89–Jun/94	
From:	Until:	<i>n</i>	Mean <i>t</i> -ratio	Posterior odds ratio	<i>n</i>	Mean <i>t</i> -ratio	Posterior odds ratio
Panel A: Close-To-Close Returns							
ann. -1 close	ann. day close	98	0.096	0.41	65	0.277	0.86
ann. day close	eff. day close				33	-0.263	-0.91
ann. day close	eff. day + 1 close	103	4.388	10.34	70	3.674	12.16
ann. day close	eff. day + 10 close	103	4.731	7.82	70	4.638	7.04
ann. day close	eff. day + 20 close	103	4.964	6.56	70	5.483	5.99
ann. day close	eff. day + 40 close	103	5.208	4.97	70	6.112	4.87
ann. day close	eff. day + 60 close	103	5.897	3.56	70	7.400	3.54
					33	2.707	1.03
					33	2.707	1.03
					33	2.707	1.03

Panel B: Returns Between Ann. and Eff. Days							
Ann. day and eff. day are same.							
ann. day close	ann. day + 1 open				70	4.372	11.69
ann. day close	ann. day + 1 open				70	-0.624	-2.35
One day between ann. and eff. days							
ann. day close	ann. day + 1 open				7	5.557	3.72
ann. day + 1 open	ann. day + 1 close				7	1.508	1.09
More than one day between ann. and eff. days							
ann. day close	ann. day + 1 open				26	2.462	5.51
ann. day + 1 open	ann. day + 1 close				26	0.603	1.31
ann. day + 1 close	eff. day close				34	2.520	2.90
eff. day - 1 close	eff. day open				26	0.229	1.42
eff. day open	eff. day close				26	1.445	2.31
eff. day close	eff. day + 1 open				33	-1.050	-2.11
eff. day + 1 open	eff. day + 1 close				33	-0.956	-3.27
Panel C: Open-To-Close Returns							
ann. day + 1 open	eff. day close				33	4.011	4.15
ann. day + 1 open	eff. day + 10 close	103	0.769	1.30	70	0.285	0.47
ann. day + 1 open	eff. day + 20 close	103	0.983	1.33	70	1.074	1.25
ann. day + 1 open	eff. day + 40 close	103	1.136	1.13	70	1.285	1.34
ann. day + 1 open	eff. day + 60 close	103	1.816	1.13	70	2.830	1.42
					33	-0.335	-0.12

5.6 Table 2, Panels B and C: Overnight, intraday, and S&P-game returns

Read.

- Under the old policy, the abnormal return from the announcement close to the next morning's open is about **4.372%**, while the open-to-close return that same day is about **-0.624%**.
- This means the old-policy premium is realized mostly **overnight**; it is not really available to someone who starts trading only after the public announcement.
- Under the new policy with more than one day between announcement and effectiveness, the next-morning overnight jump is smaller, about **2.462%**, but the stock then rises by another **2.520%** from the next day's close to the effective-day close, plus about **1.445%** from the effective-day open to the effective-day close.
- After the effective day, there is reversal: about **-1.050%** from the effective-day close to the next open and about **-0.956%** from that open to the next close.
- The abnormal return from the **open on the day after the announcement** to the **effective-day close** is about **4.011%** with a t-ratio of **4.15**.
- The average stock price in this subperiod is around **\$40**, so if one-way trading cost is below roughly **\$0.80 per share**, the average S&P-game trade remains profitable after trading costs.
- But the strategy is risky: only **25 of 33** additions produce positive abnormal returns before trading costs, and only **22 of 33** if one assumes a one-way trading cost of **\$0.25 per share**.

Economic message. The new policy created a real, tradable pre-effective-day run-up. That is the economic essence of the S&P game.

5.7 Section VI regressions: Direct evidence on the announcement rule

Read.

- The close-to-open abnormal return falls by about **1.254 percentage points** after the 1989 rule change in the regression on the post-1989 dummy.
- Within the new-policy sample, each extra day in the announcement interval reduces the overnight jump by about **0.3625 percentage points**.
- But each extra day also increases the subsequent open-to-close run-up by about **0.6147 percentage points**.

Economic message. Longer preannouncement windows smooth the opening jump but intensify the arbitrage period. The rule change reduced immediate pressure at the open, but it raised the total premium paid by indexers.

6 Short summary

- Before October 1989, S&P changes were effectively implemented overnight, and newly added stocks jumped mainly between the close and the next open.
- After October 1989, S&P preannounced additions about five days ahead to reduce opening imbalances.
- The new rule did reduce the immediate overnight jump, but it created several days for arbitrageurs to buy ahead of index funds.
- Volume and trade-size patterns show that many index funds still waited until the effective day to rebalance.
- The total run-up under the new policy was larger than under the old one, and part of it reversed after the stock entered the index.
- The evidence favors a **mostly permanent inclusion effect** plus a **temporary pre-effective-day price-pressure component** created by the new announcement rule.

7 Conclusion

7.1 Core conclusion

The paper's central conclusion is simple: **changing the timing of an announcement changed the timing and size of the price response**. Under the old policy, the inclusion premium appeared almost entirely overnight. Under the new policy, some of that premium migrated into the days between the announcement and the effective date, where it became tradable.

7.2 Economic intuition

The new rule gave the market time to respond, but that did not mean the market responded in a socially neutral way. Instead, it allowed **risk arbitrageurs** to position ahead of index funds that still cared about tracking error and therefore bought on the effective day. The result was a larger total premium paid by passive indexers, even though the next-morning imbalance was smaller.

7.3 Takeaway

This paper is a classic lesson in **market design**. A well-intentioned procedural change aimed at reducing mechanical order imbalances ended up redistributing surplus to traders who could anticipate passive demand. The broader message is that when institutional investors must trade predictably, even a small rule change can reshape prices, liquidity, and trading profits in a major way.